Supplementary Theory Proofs

Addressable state, interventions, graph sufficiency, threshold policies, sparsification, boundary placement, controlled-state reduction, and an operator-theoretic formulation

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This note collects proof surfaces for the eight items left open in the supplementary checklist. To keep the labeling restrained, only the five central structural statements are typeset as theorems; the remaining sections are presented as direct statements with proofs.

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1. A. Addressable-state necessity

Fix integers $N \geq 2$ and $k \geq 1$. In one episode sample payloads

$$V_1, \ldots, V_N \overset{\text{i.i.d.}}{\sim} \mathrm{Unif}(\{0,1\}^k).$$

Reveal the prefix $((1, V_1), \ldots, (N, V_N))$, insert a filler block longer than the local window, then draw $J \sim \mathrm{Unif}([N])$ and set the answer block to

$$Y := V_J.$$

Let $\mathcal{C}_{\mathrm{flat}}(b)$ be the class of predictors that may compress the entire prefix into a single b-bit state $C \in \{0,1\}^b$ but have no addressable external read primitive at query time. Let $\mathcal{C}_{\mathrm{obj}}$ be the class of predictors that retain the persistent object table and may directly read object J at query time. For $\pi \in \mathcal{C}_{\mathrm{flat}}(b)$, define the answer-block log-loss

$$L^{\mathrm{ans}} := \sum_{s=1}^{k} -\log_2 \hat{p}(Y_s \mid Y_{<s}, J, C).$$

Theorem. *In the one-query recall family:*

(i) every predictor $\pi \in \mathcal{C}_{\text{flat}}(b)$ satisfies

$$\mathbb{E}[L^{\text{ans}}] \geq k - \frac{b}{N};$$

(ii) there exists a predictor in $\mathcal{C}_{\text{obj}}$ using one query-time read for which

$$L^{\text{ans}} = 0 \qquad a.s.$$

Hence bounded flat state cannot achieve uniformly low loss on reusable delayed object recall, whereas persistent addressable state can.

Proof. For part (i), let C be the flat state. Because C is a measurable function of the entire prefix,

$$I(V_{1:N}; C) \leq H(C) \leq b.$$

Since the payloads are independent uniform k-bit strings,

$$H(V_{1:N}) = Nk, \qquad H(V_{1:N} \mid C) = H(V_{1:N}) - I(V_{1:N}; C) \geq Nk - b.$$

By subadditivity of conditional entropy,

$$\sum_{i=1}^{N} H(V_i \mid C) \geq H(V_{1:N} \mid C) \geq Nk - b.$$

Now J is independent of $V_{1:N}$ and C and uniform on $[N]$, so

$$H(V_J \mid J, C) = \frac{1}{N} \sum_{i=1}^{N} H(V_i \mid C) \geq k - \frac{b}{N}.$$

Because $Y = V_J$, we also have

$$H(Y \mid J, C) = H(V_J \mid J, C) \geq k - \frac{b}{N}.$$

Under log-loss, the expected sequential code length cannot beat the true posterior entropy. Therefore

$$\mathbb{E}[L^{\text{ans}}] \geq H(Y \mid J, C) \geq k - \frac{b}{N}.$$

For part (ii), store the object table

$$\mathcal{O} = \{(i, V_i)\}_{i=1}^{N}$$

throughout the prefix. When the query token J arrives, read object J once and recover V_J exactly. Since the answer block is $Y = V_J$, each next-symbol probability can be set to 1, so $L^{\text{ans}} = 0$ almost surely. □

2. B. Exact intervention semantics

Fix a finite alphabet $\mathcal{V}$ and a tombstone symbol $\perp \notin \mathcal{V}$; write $\mathcal{V}_\perp := \mathcal{V} \cup \{\perp\}$ and $\mathcal{A} := (\mathcal{V}_\perp)^N$. Let

$$E : \mathcal{A} \to \mathcal{S}, \qquad R : \mathcal{S} \times [N] \to \mathcal{P}(\mathcal{V}_\perp),$$

and state-level operators

$$D_i : \mathcal{S} \to \mathcal{S}, \qquad \text{Rep}_{i,u} : \mathcal{S} \to \mathcal{S}.$$

Let $\mathcal{S}^\dagger$ be the reachable closure generated from $E(\mathcal{A})$ under all drops and replacements. Assume that on $\mathcal{S}^\dagger$:

(1) for every s and j, $R(s,j) = \delta_{v_j(s)}$ for a unique $v_j(s) \in \mathcal{V}_\perp$;

(2) $R(D_i(s), j) = \delta_\perp$ if $j = i$ and $R(D_i(s), j) = R(s,j)$ otherwise;

(3) $R(\mathrm{Rep}_{i,u}(s), j) = \delta_u$ if $j = i$ and $R(\mathrm{Rep}_{i,u}(s), j) = R(s,j)$ otherwise;

(4) $R(E(a), j) = \delta_{a_j}$ for every $a \in \mathcal{A}$.

Let d_i and $r_{i,u}$ denote coordinatewise deletion and replacement on $\mathcal{A}$.

Theorem. *There exists a measurable map $\rho : \mathcal{S}^\dagger \to \mathcal{A}$ such that, for every $s \in \mathcal{S}^\dagger$,*

$$\begin{aligned} \rho(E(a)) &= a, \\ R(s,j) &= \delta_{\rho(s)_j}, \\ \rho(D_i(s)) &= d_i(\rho(s)), \\ \rho(\mathrm{Rep}_{i,u}(s)) &= r_{i,u}(\rho(s)). \end{aligned}$$

Thus exact history-uniform object-wise interventions force an addressable object quotient.

Proof. By assumption, for each $s \in \mathcal{S}^\dagger$ and $j \in [N]$ there is a unique value $v_j(s)$ such that

$$R(s,j) = \delta_{v_j(s)}.$$

Define

$$\rho(s) := (v_1(s), \ldots, v_N(s)) \in \mathcal{A}.$$

Since each $v_j(s)$ is the unique atom returned by the read kernel, ρ is measurable.

For encoded states, assumption (4) gives $R(E(a), j) = \delta_{a_j}$, hence $\rho(E(a))_j = a_j$ for all j and therefore $\rho(E(a)) = a$.

Read factorization is immediate from the definition of ρ:

$$R(s,j) = \delta_{\rho(s)_j}.$$

For drops, assumption (2) yields

$$R(D_i(s), j) = \begin{cases} \delta_\perp, & j = i, \\ R(s,j), & j \neq i. \end{cases}$$

Using the read factorization just proved,

$$\rho(D_i(s))_j = \begin{cases} \perp, & j = i, \\ \rho(s)_j, & j \neq i, \end{cases}$$

which is exactly $d_i(\rho(s))$.

The same argument with assumption (3) gives

$$\rho(\mathrm{Rep}_{i,u}(s))_j = \begin{cases} u, & j = i, \\ \rho(s)_j, & j \neq i, \end{cases}$$

which is precisely $r_{i,u}(\rho(s))$. □

Any substrate class that admits no such factor map ρ cannot support exact history-uniform object-wise drop and replace semantics.

3. E. Conditional graph sufficiency

Let $H_t := X_{1:t}$ and $W_t := X_{t-w+1:t}$. Let a graph state be given by a measurable map Γ_t, with readout ϕ_t, and abbreviate

$$G_t := \Gamma_t(H_t), \qquad Z_t := \phi_t(G_t).$$

Assume there exists a measurable kernel Q_t such that for every measurable set A,

$$P(X_{t+1} \in A \mid H_t) = Q_t(A \mid W_t, Z_t) \qquad \text{a.s.}$$

Assume also that the graph state is updated recurrently by a measurable rule

$$G_{t+1} = U_t(G_t, X_{t+1}) \qquad \text{a.s.}$$

Theorem. *The predictor*

$$\hat{p}_t(\cdot) := Q_t(\cdot \mid W_t, Z_t)$$

attains the full-history Bayes loss at every step:

$$\mathbb{E}[-\log \hat{p}_t(X_{t+1})] = H(X_{t+1} \mid H_t).$$

Consequently,

$$\sum_{t=1}^{T-1} \mathbb{E}[-\log \hat{p}_t(X_{t+1})] = \sum_{t=1}^{T-1} H(X_{t+1} \mid H_t).$$

Thus, on this process class, a local transducer plus graph state exactly substitutes for the global long-range part of the full-history Bayes predictor.

Proof. By assumption,

$$\hat{p}_t(\cdot) = P(X_{t+1} \in \cdot \mid H_t) \qquad \text{a.s.}$$

Log-loss is a proper scoring rule, so predicting with the true conditional law yields the conditional entropy:

$$\mathbb{E}[-\log \hat{p}_t(X_{t+1})] = \mathbb{E}[-\log P(X_{t+1} \mid H_t)] = H(X_{t+1} \mid H_t).$$

Summing over t gives the cumulative identity. The recurrent update U_t shows that the construction is online implementable. □

Approximate variant. If instead

$$D_{\mathrm{KL}}\big(P(\cdot \mid H_t) \| Q_t(\cdot \mid W_t, Z_t)\big) \le \varepsilon_t \qquad \text{a.s.,}$$

then the same predictor satisfies

$$\sum_{t=1}^{T-1} \mathbb{E}[-\log \hat{p}_t(X_{t+1})] - \sum_{t=1}^{T-1} H(X_{t+1} \mid H_t) \le \sum_{t=1}^{T-1} \varepsilon_t.$$

Proof. Conditioning on H_t and expanding the KL divergence gives

$$\mathbb{E}[-\log \hat{p}_t(X_{t+1}) \mid H_t] = H(X_{t+1} \mid H_t) + D_{\mathrm{KL}}\big(P(\cdot \mid H_t) \| \hat{p}_t(\cdot)\big).$$

The assumed KL bound therefore yields

$$\mathbb{E}[-\log \hat{p}_t(X_{t+1}) \mid H_t] \le H(X_{t+1} \mid H_t) + \varepsilon_t.$$

Taking expectations and summing over t proves the claim. □

4. G. Thresholded close/open policy

At time t, let the closable component family be

$$\mathcal{C}_t = \{C_1, \dots, C_m\}.$$

For a keep-set $A \subseteq \mathcal{C}_t$, define the accessible state

$$M_t(A) := \Big(W_t, \{X_C\}_{C\in A}, \{X_{\partial C}, \sigma(C)\}_{C\in\mathcal{C}_t\setminus A}\Big).$$

For each component C, define the one-step predictive value of keeping it open by

$$\kappa_t(C) := I\big(X_{t+1}; X_C \mid W_t, X_{\partial C}, \sigma(C)\big).$$

Assume the one-step Bayes loss decomposes additively as

$$\mathcal{L}_t(A) = \mathcal{L}_t^{\text{full}} + \sum_{C\in\mathcal{C}_t\setminus A} \kappa_t(C).$$

Theorem. *Under the additive one-step surface above:*

(i) the myopic objective

$$J_t(A) := \mathcal{L}_t(A) + \lambda \sum_{C\in A} \text{cost}(C)$$

is minimized by

$$A_t^\star = \{C \in \mathcal{C}_t : \kappa_t(C) > \lambda\,\text{cost}(C)\};$$

(ii) in the finite-horizon Bellman comparison for a fixed component C and state M,

$$Q_t^{\text{close}}(M;C) - Q_t^{\text{keep}}(M;C) = \kappa_t(C) - \lambda\,\text{cost}(C) + \Delta_t^{\text{cont}}(M;C),$$

where

$$\Delta_t^{\text{cont}}(M;C) := \mathbb{E}\big[V_{t+1}(M_{t+1}^{\text{close}}) - V_{t+1}(M_{t+1}^{\text{keep}}) \mid M_t = M\big].$$

Hence closing is optimal exactly when

$$\kappa_t(C) + \Delta_t^{\text{cont}}(M;C) \le \lambda\,\text{cost}(C).$$

Proof. For part (i), substitute the additive loss identity into the myopic objective:

$$J_t(A) = \mathcal{L}_t^{\text{full}} + \sum_{C\in\mathcal{C}_t\setminus A} \kappa_t(C) + \lambda \sum_{C\in A} \text{cost}(C).$$

Up to an additive constant,

$$J_t(A) = \sum_{C\in\mathcal{C}_t} \Big[\kappa_t(C)\,\mathbf{1}\{C \notin A\} + \lambda\,\text{cost}(C)\,\mathbf{1}\{C \in A\}\Big].$$

The optimization therefore separates componentwise. Keeping C is better than closing it exactly when

$$\lambda\,\text{cost}(C) < \kappa_t(C),$$

which proves the claimed threshold rule.

For part (ii), Bellman's equation gives

$$Q_t^{\text{keep}}(M;C) = \mathcal{L}_t^{\text{keep}}(M;C) + \lambda\,\text{memcost}^{\text{keep}}(M;C) + \mathbb{E}[V_{t+1}(M_{t+1}^{\text{keep}}) \mid M_t = M],$$
$$Q_t^{\text{close}}(M;C) = \mathcal{L}_t^{\text{close}}(M;C) + \lambda\,\text{memcost}^{\text{close}}(M;C) + \mathbb{E}[V_{t+1}(M_{t+1}^{\text{close}}) \mid M_t = M].$$

Subtracting the two identities yields

$$Q_t^{\text{close}} - Q_t^{\text{keep}} = (\mathcal{L}_t^{\text{close}} - \mathcal{L}_t^{\text{keep}}) + \lambda(\text{memcost}^{\text{close}} - \text{memcost}^{\text{keep}}) + \Delta_t^{\text{cont}}.$$

By definition, closing C instead of keeping it raises the immediate predictive loss by $\kappa_t(C)$ and lowers the maintenance cost by $\text{cost}(C)$. Hence

$$\mathcal{L}_t^{\text{close}} - \mathcal{L}_t^{\text{keep}} = \kappa_t(C), \qquad \text{memcost}^{\text{close}} - \text{memcost}^{\text{keep}} = -\text{cost}(C),$$

which gives the Bellman-corrected threshold formula. □

Surrogate extension. Assume instead that $F_t : 2^{\mathcal{C}_t} \to \mathbb{R}_{\geq 0}$ is normalized, monotone, and submodular, and impose the hard cardinality constraint $|A| \leq B$. If A_B^{gr} is the greedy keep-set and $A_B^\star$ is the size-B optimum, then

$$F_t(A_B^{\text{gr}}) \geq \left(1 - \frac{1}{e}\right) F_t(A_B^\star).$$

Proof. Let A_j be the greedy set after j steps. By submodularity and monotonicity,

$$F_t(A_B^\star) - F_t(A_j) \leq \sum_{C \in A_B^\star \setminus A_j} (F_t(A_j \cup \{C\}) - F_t(A_j)).$$

The sum on the right has at most B terms, so one of them is at least the average. Because the greedy step maximizes the marginal gain,

$$F_t(A_{j+1}) - F_t(A_j) \geq \frac{1}{B}(F_t(A_B^\star) - F_t(A_j)).$$

Rearranging,

$$F_t(A_B^\star) - F_t(A_{j+1}) \leq \left(1 - \frac{1}{B}\right)(F_t(A_B^\star) - F_t(A_j)).$$

Iterating from $j = 0$ to $B - 1$ yields

$$F_t(A_B^\star) - F_t(A_B^{\text{gr}}) \leq \left(1 - \frac{1}{B}\right)^B F_t(A_B^\star) \leq e^{-1} F_t(A_B^\star),$$

which is equivalent to the stated bound. □

5. H. Retrieval-preserving spectral sparsification

Let $K, \widetilde{K} \succeq 0$ be symmetric PSD operators on $\mathbb{R}^d$ and assume

$$(1 - \varepsilon)K \preceq \widetilde{K} \preceq (1 + \varepsilon)K \qquad \text{for some } \varepsilon \in (0, 1).$$

For a query vector q, define the dense and sparse retrieval readouts by

$$r_K(q) := Kq, \qquad r_{\widetilde{K}}(q) := \widetilde{K}q.$$

Let $D : \mathbb{R}^d \to \mathcal{P}(\mathcal{X})$ be a decoder and write

$$\ell_x(r) := -\log D(r)(x).$$

Assume that for all reachable r, r' and all $x \in \mathcal{X}$,

$$|\ell_x(r) - \ell_x(r')| \leq L_\ell \|r - r'\|_2,$$

and that the query family is norm-bounded:

$$\|q_t\|_2 \leq R \qquad \text{a.s. for all } t.$$

Theorem. *For every t,*

$$\left|\ell_{X_{t+1}}(r_{\widetilde{K}}(q_t)) - \ell_{X_{t+1}}(r_K(q_t))\right| \le L_\ell\, \varepsilon\, \|K\|_{\mathrm{op}}\, R.$$

Consequently,

$$\sum_{t=1}^{T} \mathbb{E}\big[\ell_{X_{t+1}}(r_{\widetilde{K}}(q_t)) - \ell_{X_{t+1}}(r_K(q_t))\big] \le T\, L_\ell\, \varepsilon\, \|K\|_{\mathrm{op}}\, R,$$

so the average excess log-loss is $O(\varepsilon)$.

Proof. Set $\Delta := \widetilde{K} - K$. The spectral assumption is equivalent to

$$-\varepsilon K \preceq \Delta \preceq \varepsilon K.$$

For any unit vector u,

$$|u^\top \Delta u| \le \varepsilon\, u^\top K u \le \varepsilon\, \|K\|_{\mathrm{op}}.$$

Since Δ is symmetric, the operator norm is the supremum of its quadratic form over the unit sphere, hence

$$\|\widetilde{K} - K\|_{\mathrm{op}} = \|\Delta\|_{\mathrm{op}} \le \varepsilon\, \|K\|_{\mathrm{op}}.$$

Therefore, for every query q,

$$\|r_{\widetilde{K}}(q) - r_K(q)\|_2 = \|(\widetilde{K} - K)q\|_2 \le \varepsilon\, \|K\|_{\mathrm{op}}\, \|q\|_2.$$

Applying the Lipschitz loss assumption at time t and using $\|q_t\|_2 \le R$ gives

$$\left|\ell_{X_{t+1}}(r_{\widetilde{K}}(q_t)) - \ell_{X_{t+1}}(r_K(q_t))\right| \le L_\ell\, \varepsilon\, \|K\|_{\mathrm{op}}\, R.$$

Summing over t and taking expectations proves the cumulative bound. □

Energy-only readout. If the decoder depends only on the scalar energy $a_K(q) := q^\top K q$ and the scalar log-loss is L_a-Lipschitz, then

$$|a_{\widetilde{K}}(q) - a_K(q)| = |q^\top(\widetilde{K} - K)q| \le \varepsilon\, q^\top K q.$$

Hence any uniform bound $q^\top K q \le B$ implies per-step excess loss $O(\varepsilon B)$.

Proof. The displayed quadratic-form bound is exactly the spectral approximation hypothesis written on the vector q. The Lipschitz dependence on the scalar energy transfers that bound directly to log-loss. □

Combined with any standard $(1 \pm \varepsilon)$ spectral sparsifier existence theorem for graph Laplacians or graph-induced PSD kernels, this gives the desired bridge from spectral approximation to predictive approximation.

6. Optimal write boundary placement

Let $B \subseteq \{1, \dots, T-1\}$ be a boundary set on the timeline $1, \dots, T$, and define

$$J(B) := \mathrm{Reg}_T(B) + \eta|B| \qquad (\eta > 0).$$

Suppose there is a set $\mathcal{C}^\star \subseteq \{1, \dots, T-1\}$ of predictive change points and nonnegative penalties Δ_τ such that

$$\mathrm{Reg}_T(B) = \mathrm{Reg}_T^{\mathrm{oracle}} + \sum_{\tau \in \mathcal{C}^\star \setminus B} \Delta_\tau,$$

and adding a boundary outside $\mathcal{C}^\star$ does not change the regret.

Statement. An optimal boundary set is

$$B^\star = \{\tau \in \mathcal{C}^\star : \Delta_\tau > \eta\}.$$

Ties $\Delta_\tau = \eta$ may be broken either way.

Proof. Any boundary outside $\mathcal{C}^\star$ adds the write cost η without changing the regret, so every optimum is contained in $\mathcal{C}^\star$. Restricting to $B \subseteq \mathcal{C}^\star$,

$$J(B) = \mathrm{Reg}_T^{\mathrm{oracle}} + \sum_{\tau \in \mathcal{C}^\star \setminus B} \Delta_\tau + \eta |B|.$$

Up to an additive constant,

$$J(B) = \sum_{\tau \in \mathcal{C}^\star} \Big[\Delta_\tau \, \mathbf{1}\{\tau \notin B\} + \eta \, \mathbf{1}\{\tau \in B\} \Big].$$

The optimization therefore separates pointwise. At each change point τ, inserting the boundary is better exactly when $\eta < \Delta_\tau$, which proves the formula for $B^\star$. □

Dynamic-programming form. If one writes the standard segmentation recursion

$$F(t) = \min_{0 \le s < t} \{F(s) + c(s+1, t) + \eta\}, \qquad F(0) = -\eta,$$

then under the same additive change-point surface the Bellman recursion collapses to the same threshold rule.

Proof. Under the assumption, the total objective decomposes into independent contributions from the candidate change points. The dynamic program therefore reduces to the same pointwise comparison between the omitted-change penalty Δ_τ and the write cost η. □

7. Controlled-state reduction and recurrent synthesis

Fix a finite horizon T and finite action sets $\mathcal{A}_t$. Consider the controlled objective

$$\mathcal{J}(\pi) := \mathbb{E}\left[\sum_{t=1}^{T-1} \Big(-\log q_t^\pi(X_{t+1}) + c_t^{\mathrm{mem}}(a_t, H_t) + c_t^{\mathrm{read}}(a_t, H_t) + c_t^{\mathrm{write}}(a_t, H_t) \Big) \right].$$

Let $M_t^\star = \Gamma_t(H_t)$ be the canonical predictive state. Assume that, conditional on $(M_t^\star, a_t)$, both the one-step cost law and the transition law of $M_{t+1}^\star$ are determined; equivalently, the controlled process is Markov in $M_t^\star$.

Statement. There exist measurable selectors μ_t such that the policy

$$a_t = \mu_t(M_t^\star)$$

attains the same infimum of $\mathcal{J}$ as any history-dependent optimal policy. If, in addition, $M_t^\star$ admits a separator-graph realization $\mathfrak{G}_t^\star$ with a local transition

$$\mathfrak{G}_{t+1}^\star = \mathcal{U}_t^\star(\mathfrak{G}_t^\star, X_{t+1}, a_t),$$

then an optimal policy can be implemented as the recurrence

$$a_t^\star = \mu_t(\mathfrak{G}_t^\star), \qquad \mathfrak{G}_{t+1}^\star = \mathcal{U}_t^\star(\mathfrak{G}_t^\star, X_{t+1}, a_t^\star).$$

Proof. Define the value function

$$V_t(h) := \inf_{\pi_{t:T-1}} \mathbb{E}\left[\sum_{s=t}^{T-1} \ell_s^\pi \,\middle|\, H_t = h\right],$$

where ℓ_s^π denotes the one-step total cost. At the terminal decision time, the immediate cost law depends on history only through $(M_t^\star, a_t)$, so an optimal action can already be chosen as a function of $M_t^\star$.

Assume inductively that for some measurable function $\bar{V}_{t+1}$,

$$V_{t+1}(H_{t+1}) = \bar{V}_{t+1}(M_{t+1}^\star).$$

Then the action-value at time t is

$$Q_t(h, a) = \mathbb{E}\big[\ell_t(h, a, X_{t+1}) + \bar{V}_{t+1}(M_{t+1}^\star) \mid H_t = h, a_t = a\big].$$

By the Markov sufficiency assumption, the conditional law of the quantity inside the expectation depends on h only through $M_t^\star(h)$. Hence there exists a measurable function $\bar{Q}_t$ such that

$$Q_t(h, a) = \bar{Q}_t(M_t^\star(h), a).$$

Because the action set is finite, one may choose a measurable selector

$$\mu_t(m) \in \arg\min_{a \in \mathcal{A}_t} \bar{Q}_t(m, a).$$

Then

$$V_t(h) = \min_a \bar{Q}_t(M_t^\star(h), a) =: \bar{V}_t(M_t^\star(h)),$$

so the value function itself factors through the canonical state. Backward induction shows that an optimal policy may be chosen Markov in $M_t^\star$.

If $M_t^\star$ is represented by the separator-graph state $\mathfrak{G}_t^\star$ and its update is implemented locally by $\mathcal{U}_t^\star$, the same selectors can be composed with that representation. This yields the stated separator-graph recurrence. □

8. Operator-theoretic formulation

Let

$$\mathcal{H} = \bigoplus_{e \in \mathcal{E}} \mathcal{H}_e$$

and write the predictive embedding as

$$\Phi_t(H_t) = (\phi_{e,t}(H_t))_{e \in \mathcal{E}}.$$

Let the active approximation space be

$$V_t = \bigoplus_{e \in \mathcal{E}} V_{e,t} \subseteq \mathcal{H},$$

with finite-dimensional local subspaces $V_{e,t} \subseteq \mathcal{H}_e$.

Statement 1. The orthogonal projection of Φ_t onto V_t decouples componentwise:

$$P_{V_t}\Phi_t = \big(P_{V_{e,t}}\phi_{e,t}\big)_{e \in \mathcal{E}}.$$

Proof. For any $v = (v_e)_e \in V_t$,

$$\|\Phi_t - v\|_{\mathcal{H}}^2 = \sum_{e \in \mathcal{E}} \|\phi_{e,t} - v_e\|_{\mathcal{H}_e}^2.$$

The minimization over V_t therefore separates into independent minimizations over the $V_{e,t}$. By the orthogonal projection theorem, each coordinate minimizer is $P_{V_{e,t}}\phi_{e,t}$, yielding the displayed formula. □

Choose local bases

$$\mathcal{B}_{e,t} = \{b_{e,t}^{(1)}, \dots, b_{e,t}^{(m_{e,t})}\}$$

of $V_{e,t}$ and define the coefficient vectors

$$\alpha_{e,t} := \big(\langle \phi_{e,t}, b_{e,t}^{(1)} \rangle, \dots, \langle \phi_{e,t}, b_{e,t}^{(m_{e,t})} \rangle\big).$$

Assume exact representability, namely $\phi_{e,t} \in V_{e,t}$ for all reachable histories, so that $\alpha_{e,t}$ determines $\phi_{e,t}$ exactly. Assume also neighbor-local evolution of the form

$$\phi_{e,t+1} = \Psi_{e,t}\big((\phi_{e',t})_{e' \in \mathcal{N}(e)}, X_{t+1}\big).$$

Statement 2. There exist local maps $\widetilde{\Psi}_{e,t}$ such that

$$\alpha_{e,t+1} = \widetilde{\Psi}_{e,t}\big((\alpha_{e',t})_{e' \in \mathcal{N}(e)}, X_{t+1}\big),$$

so the coefficient dynamics are graph-local.

Proof. Each coordinate of $\alpha_{e,t+1}$ has the form

$$\langle \phi_{e,t+1}, b_{e,t+1}^{(\ell)} \rangle = \left\langle \Psi_{e,t}\big((\phi_{e',t})_{e' \in \mathcal{N}(e)}, X_{t+1}\big), b_{e,t+1}^{(\ell)} \right\rangle.$$

Because exact representability allows each $\phi_{e',t}$ to be recovered from $\alpha_{e',t}$, the right-hand side is a measurable function of the neighbor coefficients and the new token X_{t+1}. Collecting those coordinate functions defines the required local map $\widetilde{\Psi}_{e,t}$. □